

# Datamatics Agentic AI Assessment

## Compliance Validator Agent — Submission Guide

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### 1. Validation Checks — What Your System Must Cover

Your agent must perform the following 10 compliance checks across all submitted invoices, covering authenticity, GST, arithmetic, TDS, and policy dimensions.

| Cat | Category                 | Check in Scope                                      |
|-----|--------------------------|-----------------------------------------------------|
| A   | Document Authenticity    | A1 — Invoice number format validation               |
|     |                          | A2 — Duplicate invoice detection across vendors     |
| B   | GST Compliance           | B1 — GSTIN format validation (15-char alphanumeric) |
|     |                          | B7 — CGST + SGST = IGST rate validation             |
| C   | Arithmetic & Calculation | C1 — Line item quantity x rate = amount             |
|     |                          | C2 — Subtotal matches sum of line items             |
| D   | TDS Compliance           | D1 — TDS applicability based on vendor type         |
|     |                          | D2 — Section determination (194C/194J/194H etc.)    |
| E   | Policy & Business Rules  | E1 — Invoice within PO amount tolerance (+/-5%)     |
|     |                          | E3 — Approved vendor list check                     |

### 2. What Your System Must Demonstrate

- Multi-agent design — at least 4 specialized agents: Extractor, Validator, Resolver, Reporter
- Process invoices in multiple formats — PDF, image, JSON, CSV
- Run all 10 checks autonomously across the full invoice batch without manual intervention
- Handle ambiguous cases with reasoning and a confidence score
- Flag any decision with confidence below 70% for human review with justification
- Log a full audit trail — agent reasoning steps, tool calls, and decision rationale per invoice

### 3. Submission Structure

Name your zip: YourFullName\_submission.zip (Example: Rahul\_Sharma\_submission.zip)

| File / Folder                   | What It Must Contain                                                                                                                |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| README.md                       | Setup and run commands — must include install steps and: <code>python src/main.py --input &lt;path&gt; --output &lt;path&gt;</code> |
| requirements.txt / package.json | All dependencies with pinned versions (e.g. <code>anthropic==0.25.0</code> )                                                        |
| .env.example                    | All environment variable names — no actual API keys                                                                                 |
| checks_manifest.json            | Declare all 10 checks with implemented status and priority (HIGH / MEDIUM / LOW)                                                    |
| src/main.py or src/index.js     | Main entry point — must accept --input and --output CLI arguments                                                                   |

| File / Folder          | What It Must Contain                                                       |
|------------------------|----------------------------------------------------------------------------|
| architecture.md        | System design — agent roles, tools used, data flow diagram                 |
| analysis.md            | Your approach, decisions made, edge cases handled                          |
| reports/ (min 3 files) | Pre-run sample outputs — one APPROVED, one REJECTED, one ESCALATE_TO_HUMAN |

#### 4. checks\_manifest.json — Required Format

Declare all 10 checks in scope with their implementation status and priority. Use this exact format:

```
{
  "candidate_name": "Rahul Sharma",
  "total_checks_implemented": 10,
  "checks": {
    "A1": { "implemented": true, "priority": "HIGH", "notes": "Regex format validation" },
    "A2": { "implemented": true, "priority": "HIGH", "notes": "Cross-invoice dedup" },
    "B1": { "implemented": true, "priority": "HIGH", "notes": "" },
    "B7": { "implemented": true, "priority": "HIGH", "notes": "" },
    "C1": { "implemented": true, "priority": "HIGH", "notes": "" },
    "C2": { "implemented": true, "priority": "HIGH", "notes": "" },
    "D1": { "implemented": true, "priority": "HIGH", "notes": "" },
    "D2": { "implemented": true, "priority": "HIGH", "notes": "" },
    "E1": { "implemented": true, "priority": "HIGH", "notes": "" },
    "E3": { "implemented": true, "priority": "HIGH", "notes": "" }
  }
}
```

#### 5. Output JSON — Required Format per Invoice

Every invoice processed must produce an output JSON in this exact structure:

```
{
  "invoice_id": "INV-2024-0001",
  "overall_decision": "APPROVED",
  "compliance_score": 96,
  "confidence": 0.95,
  "requires_human_review": false,
  "validation_results": {
    "category_a_authenticity": { "score": 2, "max_score": 2, "checks": {} },
    "category_b_gst": { "score": 2, "max_score": 2, "checks": {} },
    "category_c_arithmetic": { "score": 2, "max_score": 2, "checks": {} },
    "category_d_tds": { "score": 2, "max_score": 2, "checks": {} },
    "category_e_policy": { "score": 2, "max_score": 2, "checks": {} }
  },
  "tds_summary": {},
  "gst_summary": {},
  "audit_trail": []
}
```

overall\_decision must be one of: APPROVED | REJECTED | ESCALATE\_TO\_HUMAN | HOLD\_FOR\_VERIFICATION

## 6. Valid Output Decisions

| Decision              | When to Use                                                          |
|-----------------------|----------------------------------------------------------------------|
| APPROVED              | All 10 checks pass, no compliance violations detected                |
| REJECTED              | One or more critical checks fail — invoice must not be processed     |
| ESCALATE_TO_HUMAN     | Ambiguous or conflicting rules — agent confidence below 70%          |
| HOLD_FOR_VERIFICATION | Missing or unresolvable data — requires correction before processing |

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## 7. Instant Disqualification

Submission will be rejected immediately if any of the following apply:

- Missing README.md, checks\_manifest.json, requirements.txt, architecture.md, or analysis.md
- src/main.py or src/index.js does not accept --input and --output arguments
- Output JSON does not match the required schema
- reports/ folder has fewer than 3 sample output files
- Hardcoded decisions — mapping invoice IDs to expected results
- Single LLM call — dumping all invoice data into one prompt
- No error handling — system crashes on malformed invoice data
- Ignoring confidence — treating all decisions as 100% certain
- No Github Link as Submission